

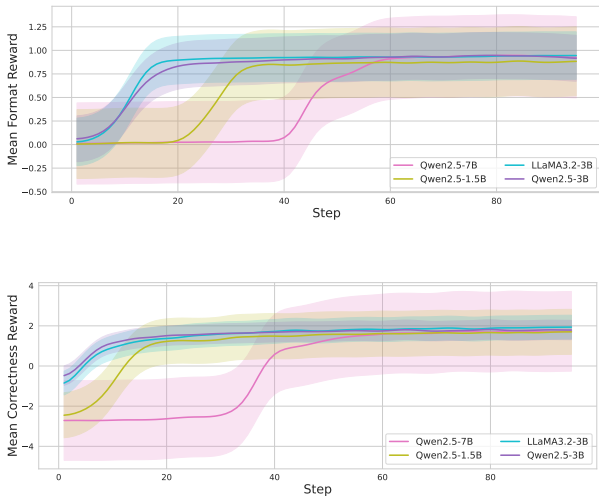
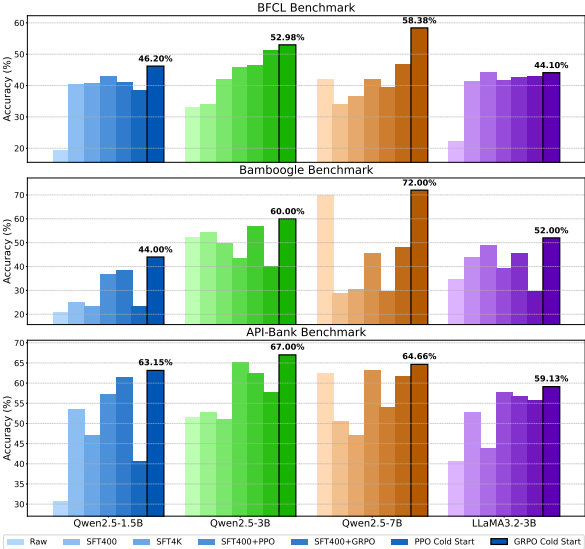


Figure 2: Main results (left) and reward trends over training steps for GRPO Cold Start across four models (right). GRPO Cold Start equipped with our proposed reward design consistently achieves the highest performance, with